

NETWORK PROJECT

Company Network Design & implementation



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2024

Company Network Design & Implementation

- **Project Overview:** The project involves designing a network for a company using **Cisco Packet Tracer**.
- **Objective:** To build a robust and secure network that connects all departments, provides continuous internet access, and ensures secure data transmission using modern techniques.

Network characteristics

The network is for a Company that is expected to have **three floors , two departments** in each floor.

• First Floor

has **Sales department**, which is expected to have **120 employees**, and **HR department** which is expected to have **120 users**.

• Second Floor

has **Finance department** which is expected to have **120 employees and Adminstrators and public relations department**, which is expected to have **120 employees**.

• Third Floor

has **ICT Department** which is expected to have **120 Employees**,
and **server room can have 12 devices**.

Network Components

1. 2 Routers Core

2. 2 Multi Layer Switches

3. At Each Floor

- Accesses Point-PC – Printer – Laptop-Switch- Tablet

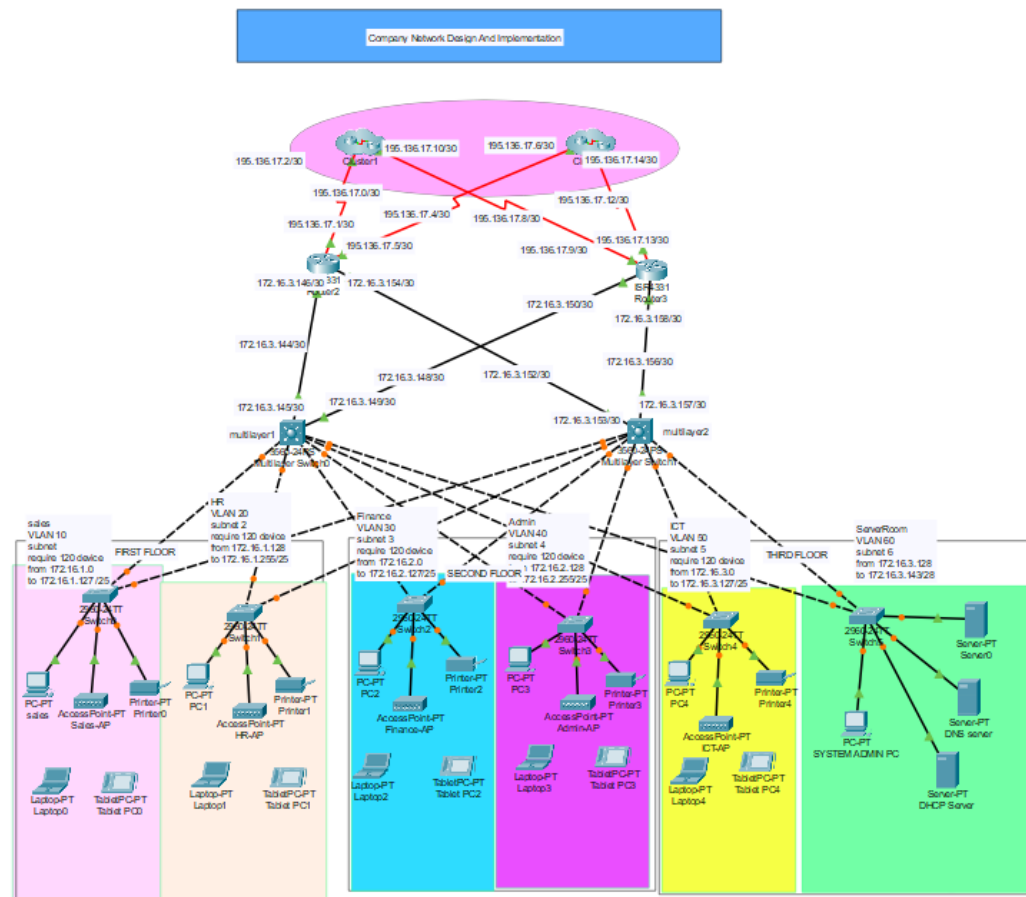
4. 2 ISP Router

5. Servers

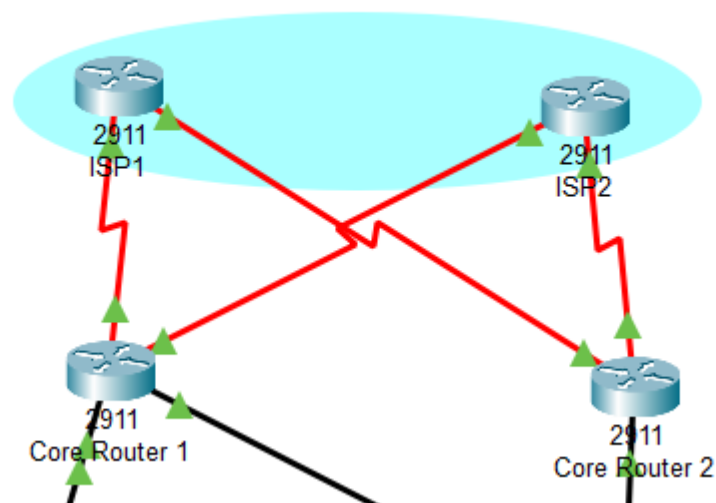
Network Requirements

1. We use Hierarchical model to provide redundancy at every layer , 2 routers and 2 multi layer switches , 2 ISPs
2. Each department is required to have wireless network and be in different VLAN and Subnetwork
3. The Base network is 172.16.1.0
4. The Company network is connected to static , public IP addresses 195.136.17.0/30 , 195.136.17.4/30,195.136.17.8/30 and 195.136.17.12/30 connected to the 2 internet provider
5. All devices in the departments are able to connected to each other

Network Topology

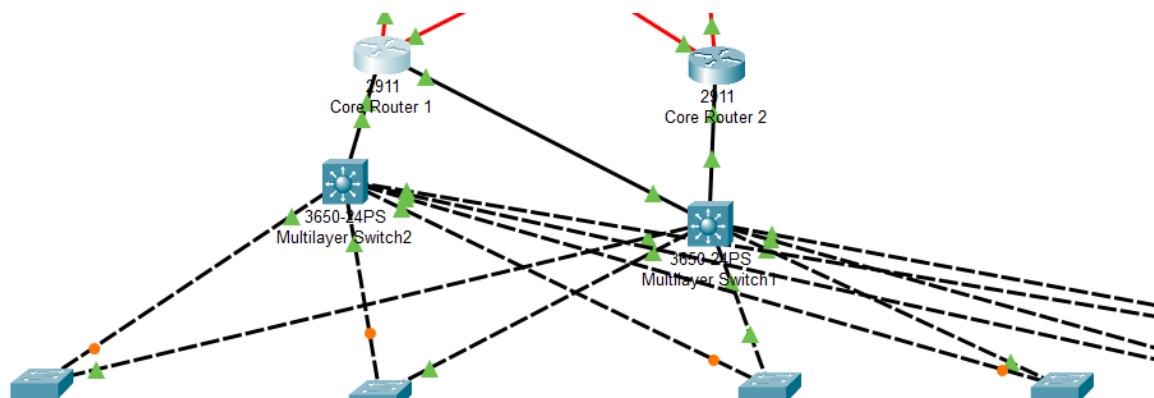


Connecting Between Routers Cores & ISPs



First we should turn off the Routers and put the HWC-2T part to make it available to Serial ports and then turn on the router then connected with each other by serial DCE. The serial DCE (Data Communications Equipment) cable is used to connect two routers via their serial interfaces.

Connecting Between Routers Cores & MultiSwitches and Switches



Copper Straight-Through Cable is used to connect different types of devices (e.g., routers to switches or PCs to switches). This cable ensures proper communication between layers in your network.

Connecting Between Switches and Department Devices

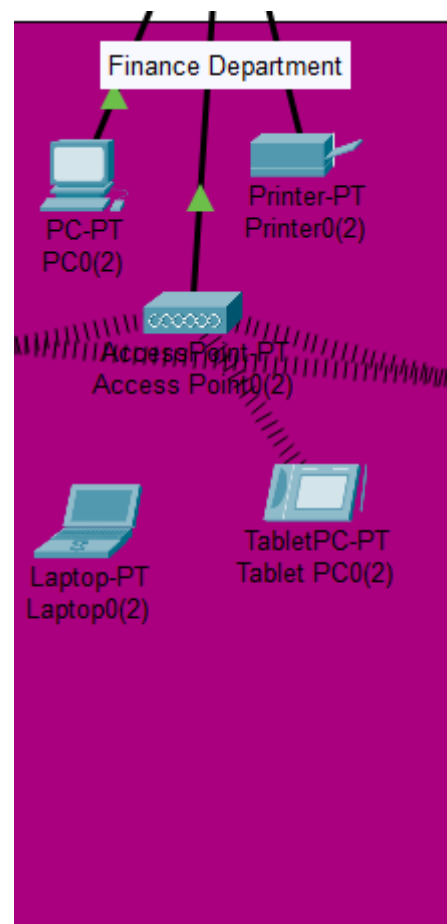
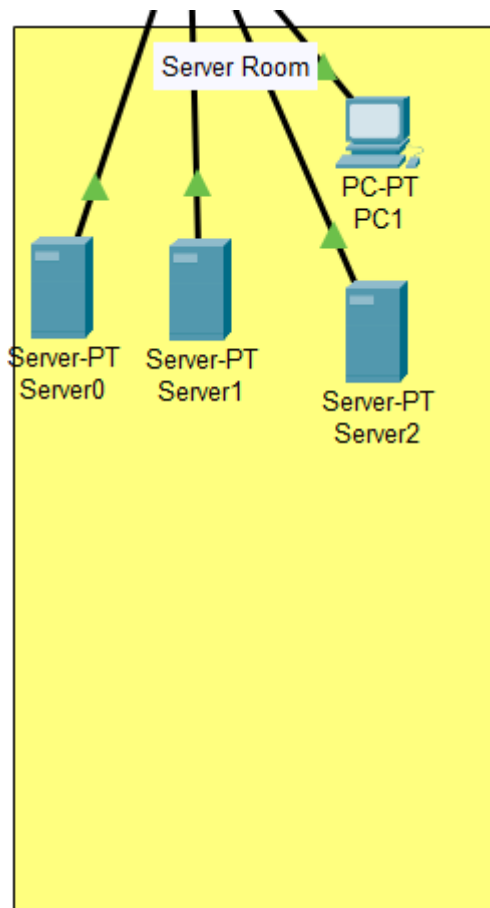
We Connect the switch to the PC , printer and The Accesses point By Copper Straight Through Cable

An **Access Point (AP)** is a networking device that allows wireless devices (like laptops, smartphones, and IoT devices)

to connect to a wired network. It acts as a bridge between the **wired network** and **Wi-Fi-enabled devices**.

The Server Room has Pc and Servers

- Servers in the server room provide centralized services required by all departments.
- Static IPs are assigned to these servers to ensure their availability and accessibility.
- They interact with routers, switches, and access points to ensure seamless operation across the entire network.



Important Step to add AC power Supply at Each Multi switch and turn on all ports in the Routers to be able to be configured

Why ?

- Ensures stable operation for routing and switching functions.
- Prevents downtime and supports network reliability.

Network Configurations

• Switch Configuration at each Department

For example the HR Department

```
HR#en
HR#config t
Enter configuration commands, one per line.  End with CNTL/Z.
HR(config)#hostname HR
HR(config)#line console 0
HR(config-line)#password cisco
HR(config-line)#login
HR(config-line)#exit
HR(config)#enable password cisco
HR(config)#no ip domain-lookup
HR(config)#banner motd #No accsses required#
HR(config)#service password-encryption
HR(config)#do wr
Building configuration...
[OK]
HR(config)#
```

Commands Brief :

1. **Changed switch name to HR** for identification.
2. **Set console and enable passwords** for security.

3. **Disabled DNS lookup** to avoid delays.
4. **Enabled password encryption** for security.
5. **Set a login banner** to warn unauthorized users.
6. **Saved the configuration** to make sure the changes are kept.

Multi Layer Switch and Core Routers Configuration

The same config of the switch plus the SSH configuration

```
Mlt-SW1(config)#ip domain name
Mlt-SW1(config)#ip domain name cisco.net
Mlt-SW1(config)#user
Mlt-SW1(config)#username admin pa
Mlt-SW1(config)#username admin password cisco
Mlt-SW1(config)#cry
Mlt-SW1(config)#crypto k
Mlt-SW1(config)#crypto key g
Mlt-SW1(config)#crypto key generate rs
Mlt-SW1(config)#crypto key generate rsa
The name for the keys will be: Mlt-SW1.cisco.net
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Mlt-SW1(config)#line vty 0 15
*Mar 1 0:8:59.705: %SSH-5-ENABLED: SSH 1.99 has been enabled
Mlt-SW1(config-line)#login loc
Mlt-SW1(config-line)#login local
```

```
Mlt-SW1(config-line)#login local
Mlt-SW1(config-line)#tran
Mlt-SW1(config-line)#transport in
Mlt-SW1(config-line)#transport input s
Mlt-SW1(config-line)#transport input ssh
Mlt-SW1(config-line)#ex
% Ambiguous command: "ex"
Mlt-SW1(config-line)#
Mlt-SW1(config-line)#exit
Mlt-SW1(config)#
Mlt-SW1(config)#
Mlt-SW1(config)#
```

Commands Brief :

1. ip domain name cisco.net

- Sets the domain name for SSH.

2. username admin password cisco

- Creates a local user admin with password cisco for SSH access.

3. crypto key generate rsa

- Generates an RSA key for SSH encryption.

4. line vty 0 15

- Configures the lines for remote access (SSH/Telnet).

5. login local

- Forces the switch to use local user credentials for login.

6. transport input ssh

- Enables only SSH for remote access.

Important Step to add SSH version for core routers and multi switches

```
Mlt-SW2(config)#ip ssh version 2
Mlt-SW2(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW2(config)#
```

• VLAN Configuration at each Department

Switch at Each Department just it varies in Vlan number and name

Sales Vlan -> 10 & name -> Sales

HR Vlan -> 20 & name -> HR

Finance Vlan -> 30 & name -> Finance

ICT Vlan -> 40 & name -> ICT

Admin Vlan -> 50 & name -> Admin

Server Vlan -> 60 & name -> Server

```
Sales(config)#int range fa0/1-2
Sales(config-if-range)#switchport mode trunk

Sales(config-if-range)#exit
Sales(config)#vlan 10
Sales(config-vlan)#name Sales
Sales(config-vlan)#exit
Sales(config)#int range fa0/3-24
Sales(config-if-range)#switchport access vlan 10
Sales(config-if-range)#exit
Sales(config)#do wr
Building configuration...
[OK]
Sales(config)#vlan 99
Sales(config-vlan)#name BlockHole
Sales(config-vlan)#exit
Sales(config)#int range gig0/1-2
Sales(config-if-range)#switchport mode access vlan 99
^
% Invalid input detected at '^' marker.

Sales(config-if-range)#switchport mode access
Sales(config-if-range)#switchport access vlan 99
Sales(config-if-range)#shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down
Sales(config-if-range)#exit
Sales(config)#do wr
Building configuration...
[OK]
Sales(config)#
```

Commands Brief :

1. int range fa0/1-2:

Selects interfaces fa0/1 to fa0/2.

2. switchport mode trunk:

Sets interfaces to trunk mode, allowing multiple VLANs.

3. vlan 10: Creates VLAN 10.

4. **name Sales:** Names VLAN 10 as "Sales".
5. **int range fa0/3-24:** Selects interfaces fa0/3 to fa0/24.
6. **switchport access vlan 10:** Assigns VLAN 10 to the selected interfaces.
7. **vlan 99:** Creates VLAN 99.
8. **name BlockHole:** Names VLAN 99 as "BlockHole".
9. **int range gig0/1-2:** Selects GigabitEthernet interfaces 0/1 to 0/2.
10. **switchport access vlan 99:** Assigns VLAN 99 to the selected interfaces.
11. **shutdown:** Disables the selected interfaces.

Security Configuration at Finance Department

```

Finance(config)#int range fa0/3-24
Finance(config-if-range)#switchport port-security maximum 1
Finance(config-if-range)#switchport port-security mac-address sticky
Finance(config-if-range)#switchport port-security violation shutdown
Finance(config-if-range)#ex
Finance(config)#
Finance(config)#
Finance(config)#do wr
Building configuration...
[OK]
Finance(config)#do sh port-security

```

Secure Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
Fa0/3	1	0	0	Shutdown
Fa0/4	1	0	0	Shutdown
Fa0/5	1	1	1	Shutdown
Fa0/6	1	0	0	Shutdown
Fa0/7	1	0	0	Shutdown
Fa0/8	1	0	0	Shutdown
Fa0/9	1	0	0	Shutdown
Fa0/10	1	0	0	Shutdown
Fa0/11	1	0	0	Shutdown
Fa0/12	1	0	0	Shutdown
Fa0/13	1	0	0	Shutdown
Fa0/14	1	0	0	Shutdown
Fa0/15	1	0	0	Shutdown
Fa0/16	1	0	0	Shutdown
Fa0/17	1	0	0	Shutdown
Fa0/18	1	0	0	Shutdown
Fa0/19	1	0	0	Shutdown
Fa0/20	1	0	0	Shutdown
Fa0/21	1	0	0	Shutdown
Fa0/22	1	0	0	Shutdown
Fa0/23	1	0	0	Shutdown
Fa0/24	1	0	0	Shutdown

```

Finance(config)#

```

Commands Brief :

1. do show port-security

- Displays the port security configuration for the device.

2. int range fa0/3-24

- Selects interfaces fa0/3 to fa0/24 for configuration.

3. switchport port-security maximum 1

- Sets the maximum allowed MAC addresses per port to 1.

4. switchport port-security mac-address sticky

- Enables sticky MAC address learning, meaning the switch will automatically learn and store the MAC addresses connected to the port.

5. switchport port-security violation shutdown

- Configures the port to shut down if a security violation occurs.

VLANs Configuration at each Multilayer Switches

```
Mlt-SW1(config)#int range gig1/0/2-7
Mlt-SW1(config-if-range)#switchport mode tr
Mlt-SW1(config-if-range)#switchport mode trunk
Mlt-SW1(config-if-range)#ex
Mlt-SW1(config)#vlan 10
Mlt-SW1(config-vlan)#name Sales
Mlt-SW1(config-vlan)#vlan 20
Mlt-SW1(config-vlan)#name HR
Mlt-SW1(config-vlan)#vlan 30
Mlt-SW1(config-vlan)#name Finanace
Mlt-SW1(config-vlan)#vlan 40
Mlt-SW1(config-vlan)#name ICT
Mlt-SW1(config-vlan)#vlan 50
Mlt-SW1(config-vlan)#name Admin
Mlt-SW1(config-vlan)#vlan 60
Mlt-SW1(config-vlan)#name Servers
Mlt-SW1(config-vlan)#ex
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW1(config)#
```

Commands Brief :

1. switchport mode tr

- **(Typo)** This command should be **switchport mode trunk**, which configures the selected interfaces to operate in **trunk mode**. Trunk mode allows the interface to carry traffic for multiple VLANs.

2. switchport mode trunk

- Configures the selected interfaces to **trunk mode**,

3. vlan 10, vlan 20, vlan 30, vlan 40, vlan 50, vlan 60

- Creates VLANs with IDs 10, 20, 30, 40, 50, and 60 on the switch.

4. name Sales, name HR, name Finance, name ICT, name Admin, name Servers

- Assigns names to the created VLANs: **Sales, HR, Finance, ICT, Admin, and Servers.**

Subnetting and IP addressing :

IP Addressing

Base Network: 172.16.1.0

First Floor

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
Sales & Marketing	172.16.1.0	255.255.255.128/25	172.16.1.1 to 172.16.1.126	172.16.1.127
HR and Logistics	172.16.1.128	255.255.255.128/25	172.16.1.129 to 172.16.1.254	172.16.1.255

Second Floor

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
Finance & Accounts	172.16.2.0	255.255.255.128/25	172.16.2.1 to 172.16.2.126	172.16.2.127
Admin & Public Relations	172.16.2.128	255.255.255.128/25	172.16.2.129 to 172.16.2.254	172.16.2.255

Third Floor

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
ICT	172.16.3.0	255.255.255.128/25	172.16.3.1 to 172.16.3.126	172.16.3.127
Server Room	172.16.3.128	255.255.255.240/28	172.16.3.129 to 172.16.1.142	172.16.3.143

Between the Routers and Layer-3 Switches

No.	Network Address	Subnet Mask	Host Address Range	Broadcast Address
R1-MLSW1	172.16.3.144	255.255.255.252	172.16.3.145 to 172.16.3.146	172.16.3.147
R1-MLSW2	172.16.3.148	255.255.255.252	172.16.3.149 to 172.16.3.150	172.16.3.151
R2-MLSW1	172.16.3.152	255.255.255.252	172.16.3.153 to 172.16.3.154	172.16.3.155
R2-MLSW2	172.16.3.156	255.255.255.252	172.16.3.157 to 172.16.3.158	172.16.3.159

Between the Routers and Layer-3 Switches

Public IP addresses 195.136.17.0/30, 195.136.17.4/30, 195.136.17.8/30 and 195.136.17.12/30

Multi switch ports Configuration to be perform as Router

Enter configuration commands, one per line. End with CNTL/Z.

```
Mlt-SW1(config)#int gig1/0/1
```

```
Mlt-SW1(config-if)#no switchport
```

```
Mlt-SW1(config-if)#ex
```

```
Mlt-SW1(config)#do wr
```

Building configuration...

Compressed configuration from 7383 bytes to 3601 bytes[OK]

[OK]

```
Mlt-SW1(config)#int gig1/0/8
```

```
Mlt-SW1(config-if)#no switchport
```

```
Mlt-SW1(config-if)#ex
```

```
Mlt-SW1(config)#do wr
```

```
-----
```

Commands Brief :

1. int gig1/0/1

- Entered configuration mode for the interface **GigabitEthernet 1/0/1**.

2. no switchport

- Changed the interface from **Layer 2** (Switchport) to **Layer 3** (Router), enabling IP-based routing.

IP addressing of Muti-Sw1:

```
Mlt-SW1(config)#int gig1/0/1
Mlt-SW1(config-if)#ip add
% Incomplete command.
Mlt-SW1(config-if)#ip address 172.16.3.145 255.255.255.252
Mlt-SW1(config-if)#no sh
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW1(config)#int gig1/0/8
Mlt-SW1(config-if)#ip address 172.16.3.149 255.255.255.252
Mlt-SW1(config-if)#no shut
Mlt-SW1(config-if)#ex
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW1(config)#
```

IP addressing of Muti-Sw2:

```
[OK]
Mlt-SW2(config)#
Mlt-SW2(config)#int gig1/0/1
Mlt-SW2(config-if)#ip address 172.168.3.153 255.255.255.252
Mlt-SW2(config-if)#no shut
Mlt-SW2(config-if)#exit
Mlt-SW2(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW2(config)#int gig1/0/2
Mlt-SW2(config-if)#ip address 172.168.3.157 255.255.255.252
Mlt-SW2(config-if)#no shut
Mlt-SW2(config-if)#ex
Mlt-SW2(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW2(config)#
```

IP addressing of Router-1:

```
Password:
CORE-R1>enable
Password:
Password:
CORE-R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CORE-R1(config)#interface GigabitEthernet0/0
CORE-R1(config-if)#ip address 172.16.3.146 255.255.255.252
CORE-R1(config-if)#no sh
CORE-R1(config-if)#ex
CORE-R1(config)#do wr
Building configuration...
[OK]
CORE-R1(config)#interface GigabitEthernet0/1
CORE-R1(config-if)#ip address 172.16.3.154 255.255.255.252
CORE-R1(config-if)#no sh
CORE-R1(config-if)#ex
CORE-R1(config)#do wr
Building configuration...
[OK]
CORE-R1(config)#
```

IP addressing of Router-2:

```
Password:
CORE-R2>enable
Password:
Password:
CORE-R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CORE-R2(config)#interface GigabitEthernet0/0
CORE-R2(config-if)#ip address 172.16.3.150 255.255.255.252
CORE-R2(config-if)#no sh
CORE-R2(config-if)#ex
CORE-R2(config)#do wr
Building configuration...
[OK]
CORE-R2(config)#interface GigabitEthernet0/1
CORE-R2(config-if)#ip address 172.16.3.158 255.255.255.252
CORE-R2(config-if)#no sh
CORE-R2(config-if)#ex
CORE-R2(config)#do wr
Building configuration...
[OK]
CORE-R2(config)#
```

IP addressing of Router-1 with ISP1 & ISP2:

The Clock Rate is the value that determines the data transfer speed between devices connected via a serial link.

```
[OK]
CORE-R1(config)#CORE-R1(config)#
CORE-R1(config)#interface Serial0/2/0
CORE-R1(config-if)#clock rate 64000
CORE-R1(config-if)#ip address 195.136.17.1 255.255.255.252
CORE-R1(config-if)#no sh
CORE-R1(config-if)#ex
CORE-R1(config)#do wr
Building configuration...
[OK]
CORE-R1(config)#interface Serial0/2/1
CORE-R1(config-if)#clock rate 64000
CORE-R1(config-if)#ip address 195.136.17.5 255.255.255.252
CORE-R1(config-if)#no sh
CORE-R1(config-if)#ex
CORE-R1(config)#do wr
Building configuration...
[OK]
```

IP addressing of Router-2 with ISP1 & ISP2:

```
[OK]
CORE-R2(config)#CORE-R2(config)#
CORE-R2(config)#interface Serial0/3/0
CORE-R2(config-if)#clock rate 64000
CORE-R2(config-if)#ip address 195.136.17.9 255.255.255.252
CORE-R2(config-if)#no sh
CORE-R2(config-if)#ex
CORE-R2(config)#do wr
Building configuration...
[OK]
CORE-R2(config)#interface Serial0/3/1
CORE-R2(config-if)#clock rate 64000
CORE-R2(config-if)#ip address 195.136.17.13 255.255.255.252
CORE-R2(config-if)#no sh
CORE-R2(config-if)#ex
CORE-R2(config)#do wr
Building configuration...
[OK]
```

IP addressing of ISP1 & ISP2:

```
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/1/0
Router(config-if)#ip address 195.136.17.2 255.255.255.252
Router(config-if)#ex
% Invalid input detected at '^' marker.

Router(config-if)#interface Serial0/1/1
Router(config-if)#ip address 195.136.17.10 255.255.255.252
Router(config-if)#do wr
Building configuration...
[OK]
```

```

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/2/0
Router(config-if)#ip address 195.136.17.6 255.255.255.252
Router(config-if)#interface Serial0/2/1
Router(config-if)#ip address 195.136.17.14 255.255.255.252
Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

```

OSPF for Routers and switches :

For multi-SW1:

```

.
4lt-SW2#config t
Enter configuration commands, one per line. End with CNTL/Z.
4lt-SW2(config)#ip routing
4lt-SW2(config)#router ospf 10
4lt-SW2(config-router)#network 172.16.1.0 0.0.0.127 area 0
4lt-SW2(config-router)#network 172.16.1.128 0.0.0.127 area 0
4lt-SW2(config-router)#network 172.16.2.0 0.0.0.127 area 0
4lt-SW2(config-router)#network 172.16.2.128 0.0.0.127 area 0
4lt-SW2(config-router)#network 172.16.3.0 0.0.0.127 area 0
4lt-SW2(config-router)#network 172.16.3.128 0.0.0.127 area 0
4lt-SW2(config-router)#network 172.16.3.128 0.0.0.15 area 0
4lt-SW2(config-router)#network 172.16.3.152 0.0.0.3 area 0
4lt-SW2(config-router)#network 172.16.3.156 0.0.0.3 area 0
4lt-SW2(config-router)#ex
4lt-SW2(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
4lt-SW2(config)#

```

Commands Brief :

- **router ospf 10:**

Activates the OSPF (Open Shortest Path First) routing protocol on the device with a process ID of 10.

- **network ... area 0:**

Defines the networks (subnets) that OSPF will advertise and include in Area 0 (the backbone area). Each network command specifies a subnet using a network address

For multi-SW2:

```
Mlt-SW1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Mlt-SW1(config)#ip routing
Mlt-SW1(config)#router ospf 10
Mlt-SW1(config-router)#router-id 2.2.2.2
Mlt-SW1(config-router)#network 172.16.1.0 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.1.128 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.2.0 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.2.128 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.3.0 0.0.0.127 area 0
Mlt-SW1(config-router)#network 172.16.3.128 0.0.0.15 area 0
Mlt-SW1(config-router)#network 172.16.3.144 0.0.0.3 area 0
Mlt-SW1(config-router)#network 172.16.3.148 0.0.0.3 area 0
Mlt-SW1(config-router)#
Mlt-SW1(config-router)#
```

For Router1:

```
CORE-R1(config)#
CORE-R1(config)#
CORE-R1(config)#router ospf 10
CORE-R1(config-router)#router-id 3.3.3.3
CORE-R1(config-router)#network 172.16.3.144 0.0.0.3 area 0
CORE-R1(config-router)#network 172.16.3.152 0.0.0.3 area 0
CORE-R1(config-router)#network 195.136.17.0 0.0.0.3 area 0
CORE-R1(config-router)#network 195.136.17.4 0.0.0.3 area 0
CORE-R1(config-router)#do wr
Building configuration...
[OK]
```

For Router2:

```
CORE-R2(config)#
CORE-R2(config)#router ospf 10
CORE-R2(config-router)#router-id 4.4.4.4
CORE-R2(config-router)#network 172.16.3.148 0.0.0.3 area 0
CORE-R2(config-router)#network 172.16.3.156 0.0.0.3 area 0
CORE-R2(config-router)#network 195.136.17.8 0.0.0.3 area 0
CORE-R2(config-router)#network 195.136.17.12 0.0.0.3 area 0
CORE-R2(config-router)#do wr
Building configuration...
[OK]
CORE-R2(config-router)#
```

For ISP Router 1 :

```
Router(config)#
Router(config)#router ospf 10
Router(config-router)#router-id 5.5.5.5
Router(config-router)#
Router(config-router)#network 195.136.17.8 0.0.0.3 area 0
Router(config-router)#network 195.136.17.0 0.0.0.3 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)# ex
05:29:41: %OSPF-5-ADJCHG: Process 10, Nbr 195.136.17.5 on Serial0/1/0 from LOADING to FULL, Loading Done
05:29:42: %OSPF-5-ADJCHG: Process 10, Nbr 4.4.4.4 on Serial0/1/1 from LOADING to FULL, Loading Done
```

For ISP Router 2 :

```
Router(config) #
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#router ospf 10
Router(config-router)#router-id 6.6.6.6
Router(config-router)#
Router(config-router)#network 195.136.17.4 0.0.0.3 area 0
Router(config-router)#network 195.136.17.12 0.0.0.3 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)# ex
```

Static IP address to ServerRoom Devices

DHCP Server

IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	172.16.3.130
Subnet Mask	255.255.255.240
Default Gateway	172.16.3.129
DNS Server	172.16.3.131

PC

☐ DHCP	☒ Static
IPv4 Address	172.16.3.132
Subnet Mask	255.255.255.240
Default Gateway	172.16.3.129
DNS Server	172.16.3.131

DNS Server

IPv4 Address	172.16.3.131
Subnet Mask	255.255.255.240
Default Gateway	172.16.3.129
DNS Server	0.0.0.0

DHCP Server Configuration

Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

FastEthernet0

 Service ☒ On ☐ Off

Pool Name

ICTPool

Default Gateway

172.168.2.129

DNS Server

172.16.3.131

Start IP Address :

172

16

2

134

Subnet Mask:

255

255

255

128

Maximum Number of Users :

120

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
AdminPool	172.168.3.1	172.16.3.131	172.16.3.6	255.255.2...	120	0.0.0.0	0.0.0.0
ICTPool	172.168.2....	172.16.3.131	172.16.2.134	255.255.2...	120	0.0.0.0	0.0.0.0
FinancePool	172.168.2.1	172.16.3.131	172.16.2.6	255.255.2...	120	0.0.0.0	0.0.0.0
HRPool	172.168.1....	172.16.3.131	172.16.1.134	255.255.2...	120	0.0.0.0	0.0.0.0
SalesPool	172.168.1.1	172.16.3.131	172.16.1.6	255.255.2...	120	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.3.128	255.255.2...	512	0.0.0.0	0.0.0.0

DNS Server Configuration

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type A Record ▾

Address

Add Save Remove

No.	Name	Type	Detail
0	www.google.com	A Record	172.16.3.131

Inter-VLAN Routing

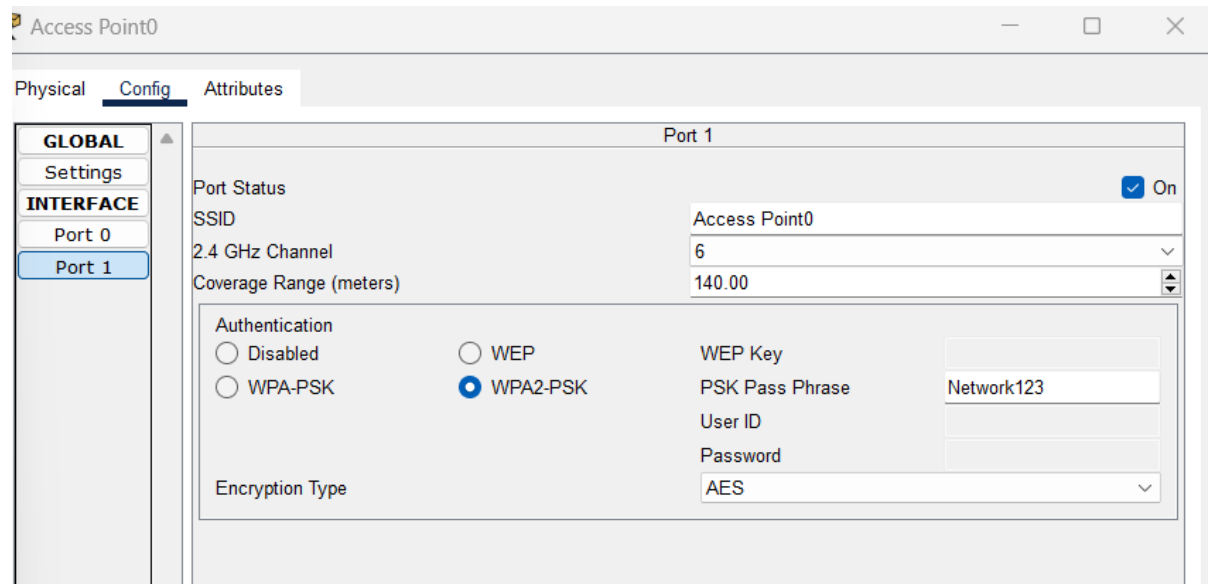
Through Multi-SW1 & Multi-SW2

```
lt-SW1(config)#int vlan 10
lt-SW1(config-if)#no sh
lt-SW1(config-if)#ip add 172.16.1.1 255.255.255.128
lt-SW1(config-if)#ip helper-address 172.16.3.130
lt-SW1(config-if)#ex
lt-SW1(config)#int vlan 20
lt-SW1(config-if)#no sh
lt-SW1(config-if)#ip add 172.16.1.129 255.255.255.128
lt-SW1(config-if)#ip helper-address 172.16.3.130
lt-SW1(config-if)#ex
lt-SW1(config)#int vlan 30
lt-SW1(config-if)#no sh
lt-SW1(config-if)#ip add 172.16.2.0 255.255.255.128
ad mask /25 for address 172.16.2.0
lt-SW1(config-if)#ip helper-address 172.16.3.130
lt-SW1(config-if)#ex
lt-SW1(config)#int vlan 40
lt-SW1(config-if)#no sh
lt-SW1(config-if)#ip add 172.16.1.129 255.255.255.128
172.16.1.128 overlaps with Vlan20
lt-SW1(config-if)#ip helper-address 172.16.3.130
lt-SW1(config-if)#ex
lt-SW1(config)#int vlan 50
lt-SW1(config-if)#no sh
lt-SW1(config-if)#ip add 172.16.3.0 255.255.255.128
ad mask /25 for address 172.16.3.0
lt-SW1(config-if)#ip helper-address 172.16.3.130
lt-SW1(config-if)#ex
lt-SW1(config)#int vlan 60
lt-SW1(config-if)#no sh
lt-SW1(config-if)#ip add 172.16.3.129 255.255.255.240
lt-SW1(config-if)#ip helper-address 172.16.3.130
lt-SW1(config-if)#ex
LINK-5-CHANGED: Interface Vlan20, changed state to up
```

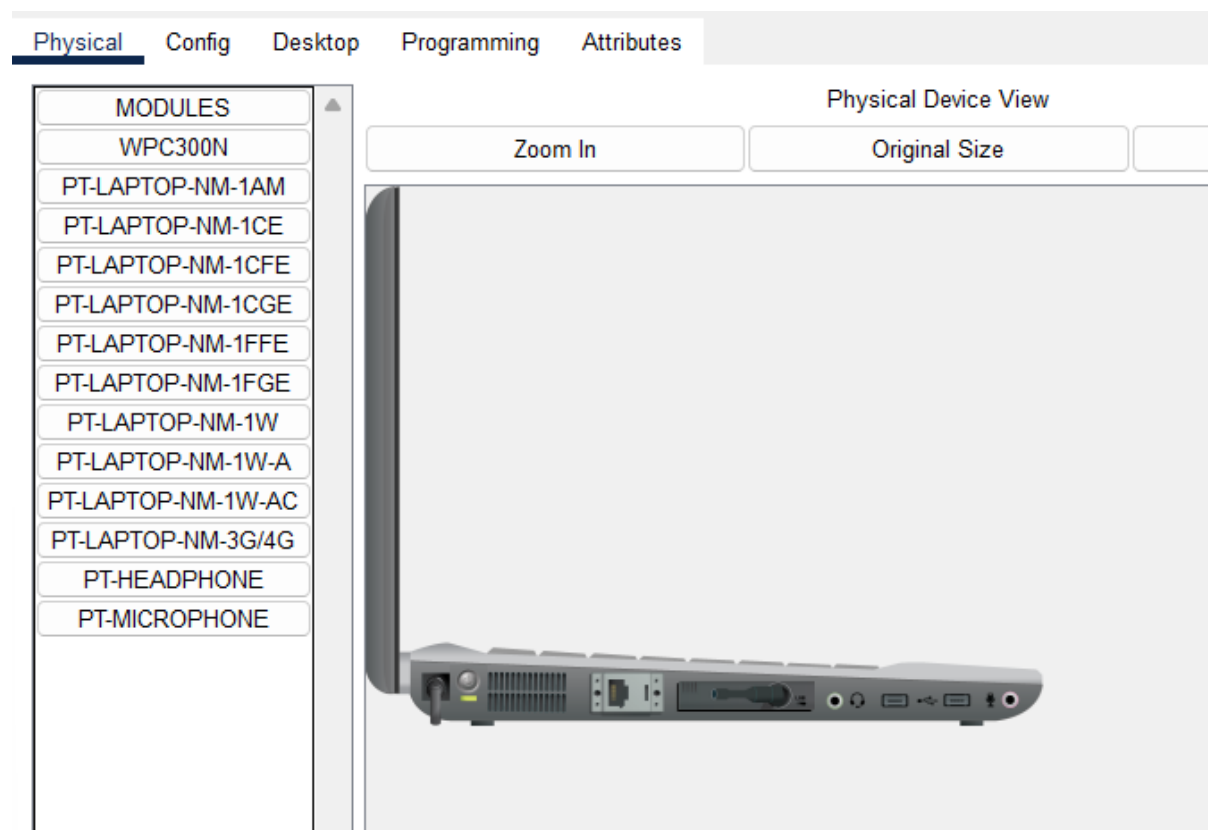
Wireless Network Configuration

For each tablet and laptop in each department

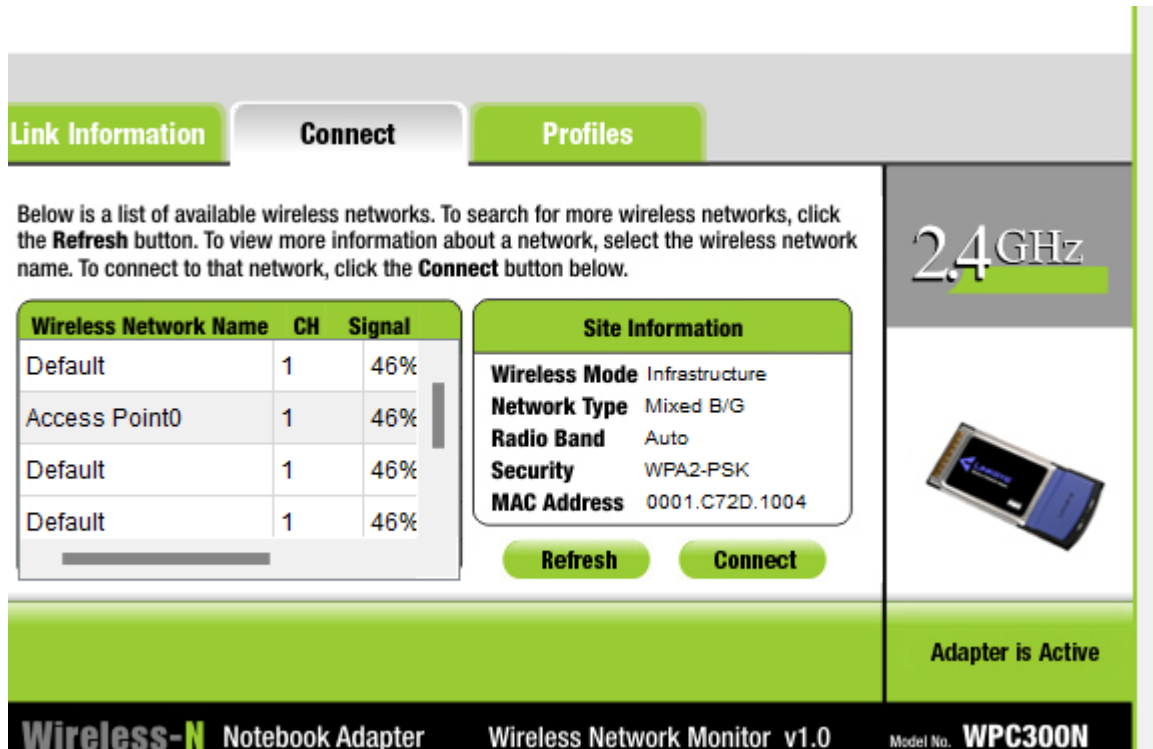
Set-up the Access point first password



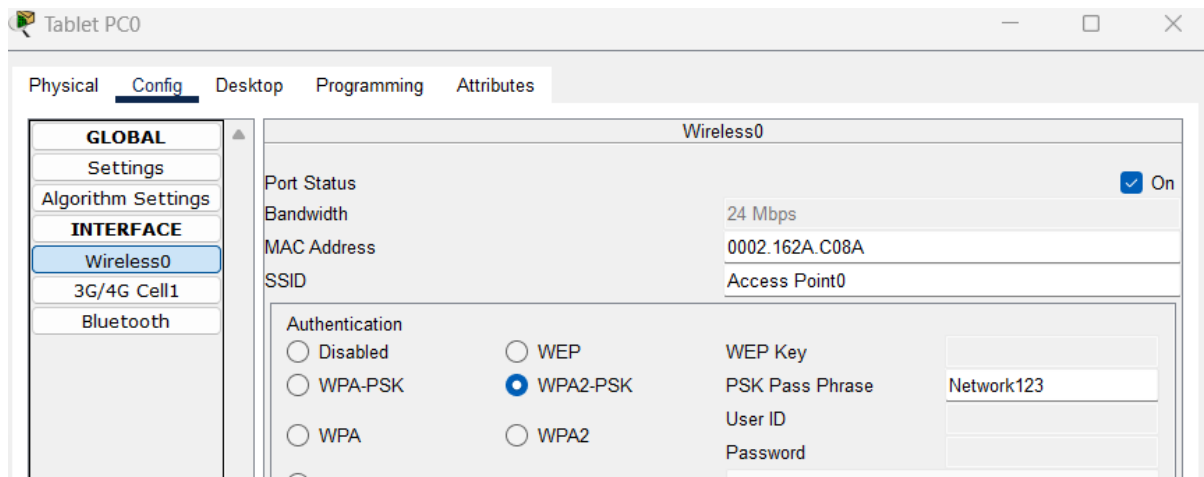
Set-up the Laptop to add WPC300N to be able connect wireless



Then Connect to the AC



Set Up the Tablet to connect to the AC



PAT and Access Control List

For Router 1

```
static specify static local-to-global mapping
CORE-R1(config)#ip nat inside source list 1 int se0/2/0 overload
CORE-R1(config)#ip nat inside source list 1 int se0/2/1 overload
CORE-R1(config)#
CORE-R1(config)#access-list 1 permit 172.16.1.0 0.0.0.127
CORE-R1(config)#access-list 1 permit 172.16.1.128 0.0.0.127
CORE-R1(config)#access-list 1 permit 172.16.2.0 0.0.0.127
CORE-R1(config)#access-list 1 permit 172.16.2.128 0.0.0.127
CORE-R1(config)#access-list 1 permit 172.16.3.0 0.0.0.127
CORE-R1(config)#access-list 1 permit 172.16.3.128 0.0.0.15
CORE-R1(config)#int range gig0/0-1
CORE-R1(config-if-range)#int nat in
CORE-R1(config-if-range)#
% Invalid input detected at '^' marker.

CORE-R1(config-if-range)#ip nat in
CORE-R1(config-if-range)#ex
CORE-R1(config)#do wr
Building configuration...

CORE-R1(config)#int se0/2/0
CORE-R1(config-if)#ip nat outside
CORE-R1(config-if)#int se0/2/1
CORE-R1(config-if)#ip nat outside
CORE-R1(config-if)#ex
CORE-R1(config)#do wr
Building configuration...
[OK]
CORE-R1(config)#
```

For Router2

```
CORE-R2>en
Password:
CORE-R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
CORE-R2(config)#
CORE-R2(config)#int gig0/0
CORE-R2(config-if)#ex
CORE-R2(config)#ip nat inside source list 1 int se0/3/0 overload
CORE-R2(config)#ip nat inside source list 1 int se0/3/1 overload
CORE-R2(config)#access-list 1 permit 172.16.1.0 0.0.0.127
CORE-R2(config)#access-list 1 permit 172.16.1.128 0.0.0.127
CORE-R2(config)#access-list 1 permit 172.16.2.0 0.0.0.127
CORE-R2(config)#access-list 1 permit 172.16.2.128 0.0.0.127
CORE-R2(config)#access-list 1 permit 172.16.3.0 0.0.0.127
CORE-R2(config)#access-list 1 permit 172.16.3.128 0.0.0.15
CORE-R2(config)#int range gig0/0-1
CORE-R2(config-if-range)#ip nat inside
CORE-R2(config-if-range)#ex
CORE-R2(config)#int se0/3/0
CORE-R2(config-if)#ip nat outside
CORE-R2(config-if)#int se0/3/1
CORE-R2(config-if)#ip nat outside
CORE-R2(config-if)#ex
CORE-R2(config)#do wr
Building configuration...
[OK]
CORE-R2(config)#
```

Commands Brief :

NAT Configuration:

- The commands `ip nat inside source list 1 int se0/2/0 overload` and `ip nat inside source list 1 int se0/2/1 overload` configure NAT to translate internal network addresses (defined in Access List 1) to the external IP addresses of interfaces `se0/2/0` and `se0/2/1` using PAT (Port Address Translation).

Access List Configuration:

- The access-list 1 permit commands define the internal subnets allowed for NAT translation:
 - `172.16.1.0/25`
 - `172.16.1.128/25`
 - `172.16.2.0/25`
 - `172.16.2.128/25`
 - `172.16.3.0/25`
 - `172.16.3.128/28`.

This ensures only the specified subnets can use the NAT configuration for external connectivity.

Default Static Route :

For Mult-SW1 and SW2

```
Mlt-SW1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Mlt-SW1(config)#ip route 0.0.0.0 0.0.0.0 gig1/0/1
Mlt-SW1(config)#ip route 0.0.0.0 0.0.0.0 gig1/0/8 70
Mlt-SW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Mlt-SW1(config)#
```

For Router one and two

```
[OK]
CORE-R2(config)#ip route 0.0.0.0 0.0.0.0 se0/3/0
CORE-R2(config)#ip route 0.0.0.0 0.0.0.0 se0/3/1 70
CORE-R2(config)#do wr
Building configuration...
[OK]
```

For Enabling DHCP for PC and Printer at each department

The screenshot shows the Cisco IOS configuration interface for a printer interface. The left sidebar has tabs for 'Physical', 'Config', and 'Attributes', with 'Config' selected. Under 'Config', there are sections for 'GLOBAL' (Settings) and 'INTERFACE' (FastEthernet0). The main area is titled 'Global Settings' and shows the 'Display Name' as 'Printer0(3)(1)'. Below this, there are options for 'Gateway/DNS IPv4': 'DHCP' (selected with a radio button) and 'Static' (unselected). There are also input fields for 'Default Gateway' and 'DNS Server'.

Testing The Network Let the PC in Sales Department connect to Pc in ICT Department

```
C:\>ping 172.16.2.154

Pinging 172.16.2.154 with 32 bytes of data:

Reply from 172.16.2.154: bytes=32 time<1ms TTL=127
Reply from 172.16.2.154: bytes=32 time=1ms TTL=127
Reply from 172.16.2.154: bytes=32 time<1ms TTL=127
Reply from 172.16.2.154: bytes=32 time=1ms TTL=127

Ping statistics for 172.16.2.154:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Testing The Network SSH

```
C:\>ssh -l admin 172.16.3.154
Could not open connection to the host, on port 22: Connect failed
C:\>ssh -l admin 172.16.3.154
```

```
Password:
```

```
CORE-R1>en
```

```
Password:
```

```
Password:
```

```
CORE-R1#
```

```
CORE-R1#
```

```
CORE-R1#
```

```
CORE-R1#
```